Resolution-driven slip inversion from InSAR line-of-sight displacement

Method, mathematics, and findings from a six-scene Venezuela study

`nisar_tools/slip`

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Abstract

This note sets out the mathematics behind the slip-inversion module in `nisar_tools` — a Python port of, and extension to, SlipSolve-Curve (x3zou, MIT) — and records what has been measured with it. The forward problem is standard: line-of-sight displacement from dislocation sources on a triangulated fault. Two pieces have no published equivalent that we could find. The first is a *per-(triangle, observation) adaptive quadrature* rule for layered elastic Green's functions, which makes a layered engine cost about the same as a closed-form homogeneous half-space and therefore removes the last computational argument for assuming a uniform crust. The second is a *joint loop* in which a single model-resolution matrix drives both the mesh refinement and the data subsampling, alternating — the published arrangement keeps these two sequential and separate.

The findings are more equivocal than the methods. Measured against matched effective rank, the mesh design, the sampling schedule and the data covariance all turn out to be second-order: the regularisation weight λ dominates the answer, four independent ways. We report that as a result rather than bury it, and quantify it: at the chosen operating point a typical slip parameter is roughly 97% regulariser, and the regularisation bias exceeds the propagated data variance by an order of magnitude at every usable λ. Scored against planted truth under measured noise, the L-curve corner that set that operating point sits at about a quarter of the effective rank the data supports, and costs 40% in model error.

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1 Scope and notation

We solve for slip on a fault surface from surface displacement measured by InSAR. Throughout, $\boldsymbol{m}$ is the parameter vector, $\boldsymbol{d}$ the data, and $\mathbf{A}$ the design matrix that maps one to the other. Vectors are columns; $'$ denotes transpose. Table 1 collects the symbols.

symbol	meaning
$\boldsymbol{d} \in \mathbb{R}^N$	line-of-sight displacement at N subsampled points
$\boldsymbol{m} \in \mathbb{R}^P$	slip parameters (two components) plus nuisance terms
$\mathbf{A} \in \mathbb{R}^{N \times P}$	design matrix: Green's functions, projected to LOS, plus ramp columns
$\mathbf{C}_d \in \mathbb{R}^{N \times N}$	data covariance, block diagonal by track
$\mathbf{L}$	discrete Laplace–Beltrami smoothing operator on the mesh
λ	regularisation weight
$\mathbf{M}$	regularised normal matrix, $\mathbf{A}'\mathbf{C}_d^{-1}\mathbf{A} + \lambda^2\mathbf{L}'\mathbf{L}$
$\mathbf{R}$	model resolution matrix
$\hat{\boldsymbol{l}}$	LOS unit vector, target to sensor
L	element size, $\sqrt{2A}$ for a triangle of area A
r	three-dimensional source–receiver separation

Table 1: Notation.

2 The forward problem

2.1 From interferometric phase to line-of-sight displacement

An unwrapped interferogram carries phase ϕ; the ground displacement projected onto the radar line of sight is

$$d = -\frac{\lambda_{\text{radar}}}{4\pi}\,\phi, \tag{1}$$

positive *toward* the sensor. The sign is not conventional decoration. With interferograms formed as ref $\cdot\ \overline{\text{sec}}$ and ISCE3's $\exp(-j4\pi r/\lambda)$ SLC phase convention, $\phi = (4\pi/\lambda)(r_{\text{sec}} - r_{\text{ref}})$, so phase *decreases* as the ground approaches the sensor and the minus sign in (1) is required.

This was wrong in every LOS product the package produced before 2026-07-28. The error was found not by a test — the synthetic fixtures built the data with $\mathbf{G}\boldsymbol{m}$ and inverted the same convention, so they were self-consistent under a global flip — but because inverting the known *dextral* San Sebastián fault recovered *sinistral* slip, with 86% of elements pinned on the polarity bound. It is now pinned by deriving the sign forward from a displacement, and a mutation test confirms the pin bites: forcing the sign positive drops the recovery test's variance reduction from $> 95\%$ to 4.9%.

2.2 Green's functions

Surface displacement from unit slip on element j, evaluated at station i, is the integral of a point-source Green's function over the element,

$$\mathbf{G}_{ij} = \int_{T_j} \boldsymbol{g}(\boldsymbol{x}_i, \boldsymbol{\xi})\, N_j(\boldsymbol{\xi})\, \mathrm{d}A(\boldsymbol{\xi}), \tag{2}$$

where N_j is the basis function (constant on the element, or a piecewise-linear "tent" peaked at node j). The projection onto LOS gives one row of $\mathbf{A}$:

$$\mathbf{A}_{ij} = \hat{\boldsymbol{l}}_i \cdot \mathbf{G}_{ij} = l_i^E \mathbf{G}_{ij}^E + l_i^N \mathbf{G}_{ij}^N + l_i^U \mathbf{G}_{ij}^U. \tag{3}$$

Two engines evaluate (2). The **homogeneous half-space** triangular dislocation element has a closed form and $\boldsymbol{g}$ never appears explicitly. The **layered** engine tabulates $\boldsymbol{g}$ on a grid of (r, z) from EDGRN and evaluates (2) by quadrature:

$$\mathbf{G}_{ij} \;\approx\; A_j \sum_{q=1}^{n_q} w_q\, \boldsymbol{g}(\boldsymbol{x}_i, \boldsymbol{\xi}_q)\, N_j(\boldsymbol{\xi}_q). \tag{4}$$

The layered engine matters because a homogeneous half-space gives the whole crust one rigidity, while the shallowest few kilometres are much softer. Assuming otherwise makes the same surface displacement require less shallow slip and more deep slip — a systematic bias in exactly the quantity a coseismic inversion is for.

2.3 Adaptive quadrature (novel)

The reference implementations use a *fixed* rule — 91 points per source–receiver pair in EDCMP's case. That includes this package's own direct ancestor: **SlipSolve-Curve** (x3zou, MIT), the MATLAB code this work is a port of and an extension to, itself descended from SlipSolve (ZeyuJin). It runs the same nine-stage arrangement — quadtree-sampled InSAR, a curved triangular mesh, a fast homogeneous TDE pass and then a layered EDGRN/EDCMP inversion — at a fixed quadrature order throughout. That is wasteful by a wide margin, because the integrand's difficulty depends entirely on how far the observation is from the element *in units of the element's own size.*

Measured on constant-slip triangular dislocations, the relative error of an order-k Gauss rule on (4) follows

$$\varepsilon \;\approx\; C\left(\frac{k\,r}{L}\right)^{-2}, \qquad C = 0.312, \tag{5}$$

with $L = \sqrt{2A_j}$ and r the full three-dimensional separation from the element centroid. The exponent is -2 (measured -2.14) because a triangle's first moment vanishes about its own centroid, so the leading error term is second order. Inverting (5) for a requested tolerance τ gives the rule actually used:

$$\boxed{k_{ij} = \left\lceil \frac{L_j}{r_{ij}}\sqrt{\frac{\sigma C}{\tau}} \right\rceil, \qquad \sigma = 4, \quad k \in \left[1,\ \sqrt{N_{\max}}\right], \quad N_{\max} = 900.} \tag{6}$$

At the default $\tau = 3 \times 10^{-3}$ this asks for $k \approx 20.4\,L/r$: order one whenever the observation is more than about twenty element-sizes away, which on a real problem is almost all of them.

Two details are load-bearing.

The safety factor σ. The fit scatter around (5) is 1.89× at one sigma, so a rule centred on the *median* leaves half of all pairs above the tolerance they asked for. Multiplying C by $\sigma = 4$ raises the order by $\sqrt{\sigma}$ and takes the fraction of pairs exceeding their tolerance from **51.7% to 5.6%**.

The near-field cap is a genuine, silent tolerance violation. Inside roughly $r \approx 2L$ the requested order exceeds $\sqrt{N_{\max}} = 30$ and is refused. The smooth $1/(kr)$ law stops describing anything once the observation is inside the element's own footprint, so the guarantee the method can honestly make is *never coarser than the fixed rule*, not *always within tolerance*. Capped pairs are now counted into the engine's provenance and warned above 5%; this is the method's one real weakness and it is reported rather than hidden.

Performance. On the six-scene problem (median $r/L = 38$), the layered engine at $\tau = 3\times10^{-3}$ uses 2.59 quadrature points per pair over 3.4 million pairs — **48 times fewer** than the 91-point reference rule — and assembles in **0.86 times** the wall time of the closed-form homogeneous engine. Against the reference rule it is 26× faster, for a slip difference of 0.47 mm rms and a correlation of 0.9999997.

Consequence. There is no longer a computational reason to prefer a homogeneous half-space. The bias it introduces is no longer being bought with anything.

3 Discretisation

3.1 Mesh and basis

The fault is a triangulated surface built from a mapped surface trace and a dip model. Slip has two components per basis function (strike-slip and dip-slip), so $P = 2n_{\text{basis}} + n_{\text{ramp}}$. Two bases are available: *element*-constant ($n_{\text{basis}} = n_{\text{elements}}$) and *node* ($n_{\text{basis}} = n_{\text{nodes}}$, piecewise-linear P1 tents). Production uses the node basis; it gives a continuous slip field with roughly half the parameters of the element basis on the same mesh.

3.2 Regularisation

Slip is regularised by a discrete Laplace–Beltrami operator on the mesh. For interior edge (i, j) the weight is the cotangent formula

$$w_{ij} = \tfrac{1}{2}(\cot\alpha_{ij} + \cot\beta_{ij})\,, \tag{7}$$

α, β being the angles opposite the edge in the two adjacent triangles, falling back to Floater's mean-value weights where the cotangent form is unsafe (obtuse or boundary configurations). Rows are assembled as $\mathbf{L}_{ii} = -\sum_j w_{ij}$, $\mathbf{L}_{ij} = w_{ij}$, then the whole operator is normalised so that its largest absolute row sum equals 2 — the scale of a plain $[+1, -1]$ difference — and repeated block-diagonally for the two slip components.

The units of the two weight families do not match, and mixing them was a real bug. Cotangents are dimensionless; mean-value weights carry 1/length. On a metre-scale mesh the ratio was measured at 2397, which is the median edge length — so a single node that fell back to the mean-value branch received a smoothing row three orders of magnitude weaker than its neighbours and was effectively unregularised. The symptom was one small patch of anomalous slip in an otherwise clean model. The fix multiplies the mean-value fallback by the edge length and a scale factor of $\frac{1}{2}$. It required *both* a fully starved node *and* a free-surface position to fire, which is why it survived so long.

3.3 Nuisance parameters

Each track carries an unknown reference level and a long-wavelength ramp (orbital error, residual troposphere and ionosphere). These are solved *jointly* with slip rather than removed beforehand, as extra columns of $\mathbf{A}$: a polynomial of chosen degree in the local (x, y) of each track. The production choice is quadratic, six columns per track, 36 columns over six scenes.

Solving them jointly matters. Pre-removing a ramp by fitting it to the data subtracts real deformation wherever the deformation is long-wavelength or reaches the frame edge; solving jointly lets the data decide the split. It is not free — Section 7.4 measures what the ramp absorbs even when solved jointly — but it is the lesser error.

Do not merge scenes into one raster before inverting. A mosaicked raster has *one* arbitrary constant and *one* ramp, so a second scene's independent nuisance is absorbed as fictitious slip. Measured on a controlled synthetic with a planted right-lateral patch: two tracks concatenated as separate observation sets recovered peak slip 1.587 m against a truth of 1.626 m (correlation 0.979, moment ratio 1.001), while the same data mosaicked into one raster returned 5.678 m — 3.5× the truth (correlation 0.553, moment ratio 1.671). **Variance reduction stayed at 98.4%** for the wrong answer, so the usual goodness-of-fit statistic does not catch this.

4 The inverse problem

4.1 Data covariance

InSAR noise is spatially correlated — atmosphere, not shot noise — so the independent-errors assumption is wrong at exactly the scales slip inversion cares about. We estimate an isotropic exponential covariance per track from the far-field residual after removing a degree-2 trend:

$$\gamma(h) = n + s\left(1 - e^{-h/a}\right), \qquad C(h) = s\, e^{-h/a} + n\,\delta(h), \tag{8}$$

with sill s, range a and nugget n fitted to the empirical semivariogram. The full $\mathbf{C}_d$ is block diagonal by track, since separate acquisitions have independent atmospheres.

The generalised least-squares problem is then whitened by the Cholesky factor $\mathbf{C}_d = \boldsymbol{F}\boldsymbol{F}'$, applied blockwise:

$$\tilde{\mathbf{A}} = \boldsymbol{F}^{-1}\mathbf{A}, \qquad \tilde{\boldsymbol{d}} = \boldsymbol{F}^{-1}\boldsymbol{d}, \tag{9}$$

by triangular solve rather than by forming $\mathbf{C}_d^{-1}$. Block diagonality is what makes this affordable: six blocks of ∼1000–1600 at a few MB each, against 784–909 MB for a dense $\mathbf{C}_d$ at production size.

4.2 The solve

The estimate minimises

$$\boxed{\hat{\boldsymbol{m}} = \arg\min_{\boldsymbol{m}\in\mathcal{B}} \left\|\boldsymbol{F}^{-1}\left(\mathbf{A}\boldsymbol{m} - \boldsymbol{d}\right)\right\|_2^2 \;+\; \lambda^2 \left\|\mathbf{L}\boldsymbol{m}\right\|_2^2} \tag{10}$$

subject to box constraints $\mathcal{B}$: a *polarity* constraint fixing the sign of the strike-slip component to the known sense of the fault, and magnitude bounds on each component. The bounded problem is solved by `scipy.optimize.lsq_linear`.

The unconstrained normal matrix,

$$\mathbf{M} = \mathbf{A}'\mathbf{C}_d^{-1}\mathbf{A} + \lambda^2\mathbf{L}'\mathbf{L}, \tag{11}$$

is what every resolution and uncertainty quantity below is built from. It is Cholesky-factorised once per λ and reused.

The bounds interact with everything downstream and must be checked, not assumed. All the linear analysis in Sections 5 and 6 describes the *unbounded* problem. It is trustworthy only when few parameters sit on a bound. On the current production model 6 of 374 nodes (1.6%) are active, all on the polarity constraint — but an earlier run, before the LOS sign fix, had **86%** pinned, and there the linearisation would have been meaningless. This is a property of the model, not of the code, and is re-checked per run.

5 Resolution analysis

5.1 Model and data resolution

With $\mathbf{M}$ from (11), the model resolution matrix is

$$\mathbf{R} = \mathbf{M}^{-1}\mathbf{A}'\mathbf{C}_d^{-1}\mathbf{A}, \tag{12}$$

so that $\hat{\boldsymbol{m}} = \mathbf{R}\,\boldsymbol{m}_{\text{true}}$ in the noiseless unbounded limit. The data resolution (influence) matrix is

$$\boldsymbol{N} = \mathbf{A}\mathbf{M}^{-1}\mathbf{A}'\mathbf{C}_d^{-1}, \tag{13}$$

and the two satisfy

$$\operatorname{tr}(\mathbf{R}) = \operatorname{tr}(\boldsymbol{N}) \;\equiv\; \rho(\lambda), \tag{14}$$

the **effective rank**: the number of parameters the data actually pays for. Identity (14) is used as a runtime assertion — it holds only if both sides whiten the same $\mathbf{A}$, so a missed whitening site fails loudly instead of silently biasing the analysis.

Effective rank, not λ, is the right axis for comparing configurations. Two meshes at the same λ are damped by different amounts; at the same ρ they are not. Every comparison in Section 7 is made at matched ρ for this reason.

5.2 Averaging kernels

Row k of $\mathbf{R}$ is the averaging kernel for parameter k: it says which neighbourhood of the true field the estimate at k actually reports. Its width is summarised by second moments of $|\mathbf{R}_{k,:}|$ in the two fault-surface directions separately, giving anisotropic along-strike and down-dip resolving lengths.

These second moments are not Barnhart & Lohman's metric, despite serving the same purpose. They fit a *Gaussian* to row k and take its 1σ. The averaging kernel here is strongly non-Gaussian: measured on the six-scene model, **19.4% of a row's $|\mathbf{R}|$ mass sits beyond 3σ of the fitted Gaussian**, against the 0.27% a Gaussian would put there — a 70× excess. Second moments weight by distance squared, so they read those tails and come out **3.09**× the Gaussian σ along strike (39.3 km against 11.5 km). Trimming the weakest 10% of each row brings the ratio to 1.05, which identifies the tails as the entire difference.

5.3 Resolution-driven meshing

Two criteria are implemented for turning resolution into a target element size.

Diagonal (default). Even out diag($\mathbf{R}$) across the mesh:

$$h_{\text{new}} = h \left(\frac{\bar{r}}{r}\right)^{g}, \qquad g = 0.5, \tag{15}$$

with $\bar{r}$ the median resolution, clipped to a factor of 3 per step and smoothed over the mesh. Under-relaxing ($g < 1$) is deliberate: resolution is itself a function of the discretisation, so a full step chases its own tail.

Kernel width. Set $h = f\times$ (kernel width), $f = 0.5$: two elements per resolvable feature, the sampling analogue of Nyquist. This is the closer analogue of the published method.

Element *count* is held separately by an absolute integer budget, because the size field sets the *distribution* and not the scale. Targeting the median looks self-normalising and is not: the resolution distribution is heavily skewed, most elements hit the growth cap while few can shrink, and the mean element size climbs every round. Measured without a budget, a loop coarsened 330 elements to 186, then 46, then 44, with variance reduction collapsing $94.3\% \to 91.6\% \to \mathbf{0.74}\%$. It ran to completion, converged, and returned nothing.

5.4 The joint loop (novel)

Barnhart & Lohman (2010) size a mesh from resolution; Lohman & Simons (2005) subsample data from a predicted field. Their papers are explicit that the two stay sequential and separate. The loop here drives *both* from the same $\mathbf{R}$, alternating:

1. Subsample each scene on the current model plus resolution field.
2. Assemble, solve, factorise $\mathbf{M}$ once.
3. Re-size elements from that $\mathbf{R}$; rebuild the mesh at fixed budget.
4. Repeat; hold the mesh fixed on the final round so the convergence measure is exact.

Holding the mesh on the last round is necessary, not cosmetic. Re-meshing perturbs a slip field by 0.1–1 m at element boundaries whatever the model does — two models fitting the *same* data on 240 and 232 elements agree to a 2 mm median and still differ by 0.97 m at the worst probe point — so a centimetre tolerance is unreachable while the mesh is moving.

6 Uncertainty

In a regularised inversion the error has two parts that trade off along the L-curve, and reporting only one is misleading.

6.1 Variance

Propagating data noise through the linearised estimator gives

$$\mathbf{C}_m = \mathbf{M}^{-1}\mathbf{A}'\mathbf{C}_d^{-1}\mathbf{A}\,\mathbf{M}^{-1}. \tag{16}$$

When the estimator's weighting does not match the true noise — which is the production case, since the shipped model is solved with plain weights while the noise is correlated — the correct expression is the *sandwich*

$$\boxed{\mathbf{C}_m = \mathbf{M}^{-1}\,\mathbf{A}'\mathbf{W}\mathbf{C}_d\mathbf{W}\,\mathbf{A}\,\mathbf{M}^{-1}} \tag{17}$$

with $\mathbf{W}$ the diagonal weight matrix the solve actually used. Note $\mathbf{W}$ appears *twice* on each side; applying it once understates the result by a constant factor (measured: 28.78× too large in the first implementation, with the correlation against Monte Carlo unchanged at 0.978, which is what identified it as a pure scale error rather than a structural one).

6.2 Bias

The regulariser's pull on the model is

$$\boldsymbol{b} = (\mathbf{R} - \boldsymbol{I})\,\boldsymbol{m}, \tag{18}$$

and $1 - \mathbf{R}_{kk}$ is directly the fraction of the estimate at k that came from smoothing rather than from data.

Equation (18) must be evaluated at a *fixed* reference model, not at the recovered model. Evaluated at $\hat{\boldsymbol{m}}(\lambda)$ the bias appears to *fall* with λ (measured $0.317 \to 0.098$) simply because the recovered model shrinks; and even the relative bias is non-monotonic ($0.203 \to 0.431 \to 0.290$) because $\hat{\boldsymbol{m}}$ rotates toward the directions $\mathbf{R}$ preserves. Only against a fixed reference does the curve mean what it appears to mean.

6.3 Monte Carlo

The honest check draws realisations of correlated noise from the fitted $\mathbf{C}_d$, adds them to the data, and re-solves through the *actual bounded* solver, which captures the bounds and the nonlinearity the linearisation discards. The Green's matrix is reused across realisations, so this is N solves and not N assemblies.

Not bootstrap. A naive observation resample assumes independence, which the measured 22–33 km correlation length says is false, and a block bootstrap would need blocks larger than that — leaving too few independent blocks in a 5700-sample set. Drawing from a covariance we actually measured is strictly better here.

7 Findings

7.1 The noise structure of the six scenes

Table 2 gives the fitted exponential covariances. The striking result is the third column.

scene	sill (mm)	nugget (mm)	range (km)
D135	13.4	0.42	31.3
D134	16.3	0.52	32.6
A061	18.1	1.19	22.5
A162	20.7	2.25	22.7
D126	27.3	3.27	23.2
D054	40.9	1.29	29.3

Table 2: Exponential covariance fitted to each scene's far-field residual after a degree-2 detrend. Sill and nugget are quoted as standard deviations.

The sill varies by a factor of three across scenes and the nugget by nearly eight, but the **correlation range is 22–33 km on all six** — across two different processing chains (NISAR GSLC and ALOS). That is a property of the atmosphere over this region, not of the processing, and it is the number that makes a block bootstrap infeasible.

7.2 The data covariance does not improve the slip estimate

This was the load-bearing gap identified before the work started, and closing it produced a negative result. Against a synthetic with known truth, over five seeds, the whitened solve recovered slip at **0.147 m rms error** against **0.069 m** for the plain-weighted solve — worse, consistently, 5 seeds out of 5.

The reason is not a bug. Generalised least squares is minimum-variance only among *unbiased* estimators. This estimator is neither: it is bound-constrained and, at the operating point, roughly 97% regulariser. Whitening correctly down-weights spatially redundant information, but what it hands back to the regulariser is a weaker data constraint, and the regulariser is the larger term.

What $\mathbf{C}_d$ *is* good for is honest error bars — the sandwich (17) needs it — and a χ^2 per degree of freedom that means something. It is retained for those, not for the point estimate.

7.3 The element budget and λ

The inherited budget (1696 elements, 1836 slip parameters) traced back to a default edge length, not to a measurement. Against it, $\mathrm{tr}(\mathbf{R}) = 38.7$: the data paid for about 1 parameter in 47.

Re-measured, the L-curve corner sits at effective rank ≈ 32 on the 1696-element mesh and ≈ 21 on the 651-element unstructured mesh. The budget was **11–57 times larger than the data supports**. Production now runs 660 elements (748 parameters), which reaches rank 41.5 at $\lambda = 30$ and so clears the corner with headroom, while 416 elements caps at 25.7 and cannot.

7.4 The ramp absorbs signal, and how much depends on λ

Ramp degree buys fit monotonically (variance reduction $96.50 \to 99.55 \to 99.66 \to 99.71$ for offset / linear / quadratic / cubic) while moment changes by $1.00 \to 1.56 \to 1.65 \to 1.48$ relative to the offset-only case. Fit alone cannot choose the degree.

The synthetic recovery experiment measures directly what the ramp eats, because truth is known and the data are noiseless: any ramp coefficient returned is signal the ramp has absorbed. On the 40 km magnitude checkerboard the maximum absorbed amplitude falls by a factor of ten as damping is relaxed (Table 3). **Ramp leakage is largely a symptom of over-damping**: when the slip parameterisation cannot fit the data, the ramp absorbs the misfit.

		pattern correlation			max |ramp| (m)		
rank	λ	20 km	40 km	60 km	20 km	40 km	60 km
9.7	388	0.018	0.278	0.250	0.730	0.670	0.728
20.6	144	0.033	0.437	0.423	0.471	0.392	0.443
39.8	50	0.070	0.615	0.634	0.250	0.167	0.195
79.1	13	0.168	0.726	0.751	0.118	0.070	0.079

Table 3: Recovery of magnitude checkerboards against effective rank, on noiseless synthetic data with real coverage and the unstructured layout. Correlation is computed on the oscillating part, after removing each field's own mean. Production operates at rank ≈ 21, the second row. λ is quoted for the 40 km case; the other two differ by under 1%.

Three things fall out.

There is a resolution cliff between 20 and 40 km. The 20 km checkerboard **never recovers**: correlation runs from 0.018 to 0.168 across an eight-fold change in effective rank, and even at rank 79 the recovered field (Figure 1, top row) bears no resemblance to the target. Relaxing λ does not buy it back, so this is a genuine information limit and not a regularisation choice.

Above that cliff, wavelength stops mattering and damping takes over. The 40 and 60 km cases track each other to within 0.02 in correlation at every rung — 0.28/0.25, 0.44/0.42, 0.62/0.63, 0.73/0.75 — while each varies by a factor of 2.6 across the ladder. Between 40 and 60 km the answer is set by λ, not by the target.

The kernel width marks "half-recovered", not "recovered". At the production operating point the measured along-strike kernel width is ≈ 40 km, and a 40 km checkerboard comes back at correlation 0.44. So the width is a fair marker of where recovery becomes partial, and an optimistic one if read as where recovery becomes clean.

A caution on metrics: at 20 km the *amplitude ratio* sits at 0.88 while the correlation is 0.03. The model carries very nearly the right amount of variance and puts almost none of it in the right place. Amplitude ratio alone is not a recovery statistic.

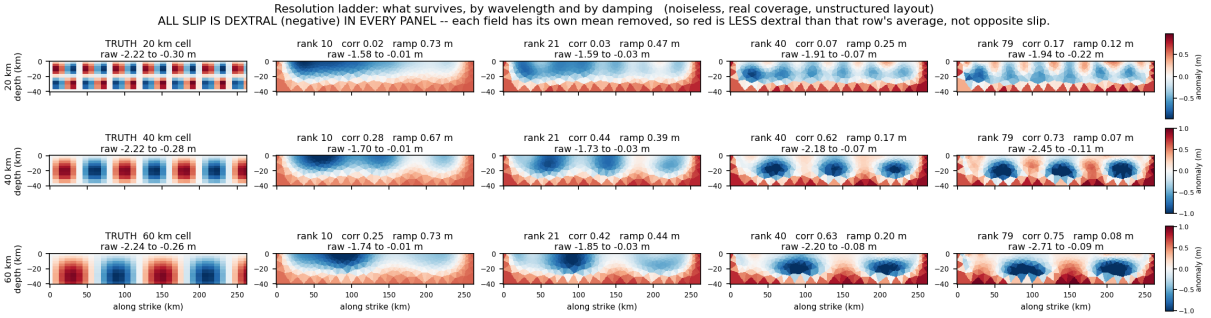


Figure 1: The resolution ladder. Rows are planted wavelength, columns are effective rank; the leftmost column is truth. Same noiseless data, same estimator, same mesh budget — only λ varies along a row. The 20 km target never resolves; the 40 and 60 km targets are limited by damping rather than by wavelength. Each panel has its row's colour scale, with the uniform dextral base removed.

7.5 Adaptive quadrature

Reported in Section 2.3: 48× fewer quadrature points than the reference rule, layered assembly at 0.86× the cost of the closed-form homogeneous engine, median relative error 7.2×10^{-4} against real Crust2.0 tables. This is the cleanest positive result in the study.

7.6 The joint loop is an efficiency method, not an accuracy method

Four arms at one budget, one round count, compared at matched effective rank: a fixed uniform mesh (null), resolution-sized once then sampled separately (the published arrangement), the joint loop, and the joint loop with the kernel-width metric. The joint loop gains **3.6% over the null** at the corner rank — and costs **32% more**.

The earlier belief that it was also cheaper was an artefact of a defect: the loop was performing seven Cholesky factorisations of a $\sim 1870^2$ matrix per round where one suffices. With that fixed, the cost advantage disappears.

The kernel-metric arm *lost* at every effective rank, including to a fixed mesh at low rank. Section 5 explains why, and the explanation exonerates the published method: our kernel width is a second moment, not their Gaussian fit, and the reduction `min` selects the down-dip width on 100% of parameters, so the along-strike number never reaches the size field at all. This is evidence against *this estimator of the kernel*, not against Barnhart & Lohman's method, which has not been run here.

7.7 Uncertainty on the production model

The analytic sandwich (17) reproduces a 40-draw Monte Carlo through the bounded solver at a median ratio of **1.15** and a correlation of **0.79** over 374 nodes, in under a second against roughly 40 minutes. The shipped model has peak slip $\mathbf{2.28 \pm 0.07}$ m, with a mean σ of 0.035 m analytically and 0.028 m by Monte Carlo.

These are variance only, and variance is the smaller half. The regularisation bias (18) exceeds the propagated data variance by an order of magnitude at every usable λ. An error bar quoting only σ would be badly over-confident. The correct presentation is the bias–variance trade-off against λ, not a single number: moving λ moves both, so they cannot be added into one interval.

7.8 Synthetic recovery, and a test that measured its own bound

Truth was planted on the mesh, forward-modelled through the real look geometry and coverage masks, and inverted with the production configuration. Noiseless by design: a checkerboard's adjacent lobes cancel at the surface, leaving only 10–18 mm rms of signal against a fitted atmosphere of 12–41 mm, so noise would swamp it and the test would report "nothing recovers".

The first suite was invalid, and every summary statistic stayed plausible while it was. The textbook checkerboard alternates sign; production runs a polarity constraint forcing strike-slip ≤ 0. **43.8% of the planted model was in a region the solver is forbidden to enter.** Every recovered model came back single-signed at all 18,000 probe points, with 25–32% sitting at exactly zero (Figure 2). The scores were measuring the bound.

The fix keeps the estimator and changes the target: alternate the *magnitude* of dextral slip rather than its sign,

$$s(\boldsymbol{x}) = -\left[\, b + c\, \sin(\pi x_s/\ell)\, \sin(\pi x_z/\ell) \,\right], \qquad b = 1.25,\ c = 1.0, \tag{19}$$

spanning -0.25 to -2.25 m — a factor of nine between weak and strong cells, entirely inside the bound. Sixteen inversions were run on it: three cell sizes, two layouts at 40 km, each at four points on the rank ladder (Table 3, Figure 1).

Two conclusions survive the confound, because the bound applied identically to every arm:

- **Mesh layout is not the lever, at any damping.** Row-structured and unstructured layouts on the 40 km target agree to within 3% at every rung of the ladder — correlation 0.278/0.278, 0.422/0.437, 0.610/0.615, 0.729/0.726 — and the sign of the difference *flips* at the last one, so it is scatter and not a trend. On real data, compared only to each other, the same two layouts differ by 0.78 m locally.
- **The quadratic ramp absorbed 0.38–0.39 m** of a ± 1.94 m checkerboard under real coverage, and 0.14–0.16 m under full coverage. Coverage gaps roughly double what the ramp takes.

A separate check confirms the mesh is not the limiting factor: projected onto the 660-element mesh's nodes, the planted checkerboards retain 0.993, 0.999 and 0.993 of their amplitude at 20, 40 and 60 km cell size. Any amplitude loss in the recovery is the regulariser, not the discretisation.

7.9 The L-curve corner is not where the error is smallest

The ladder above is noiseless, so it measures only the *bias* half of the error. That is exactly the half the L-curve corner is not trying to minimise, so the two must be compared under noise before either can be believed. Five independent realisations were drawn from the covariances of Table 2, added to the 40 km target, and inverted over six rungs (Table 4, Figure 3).

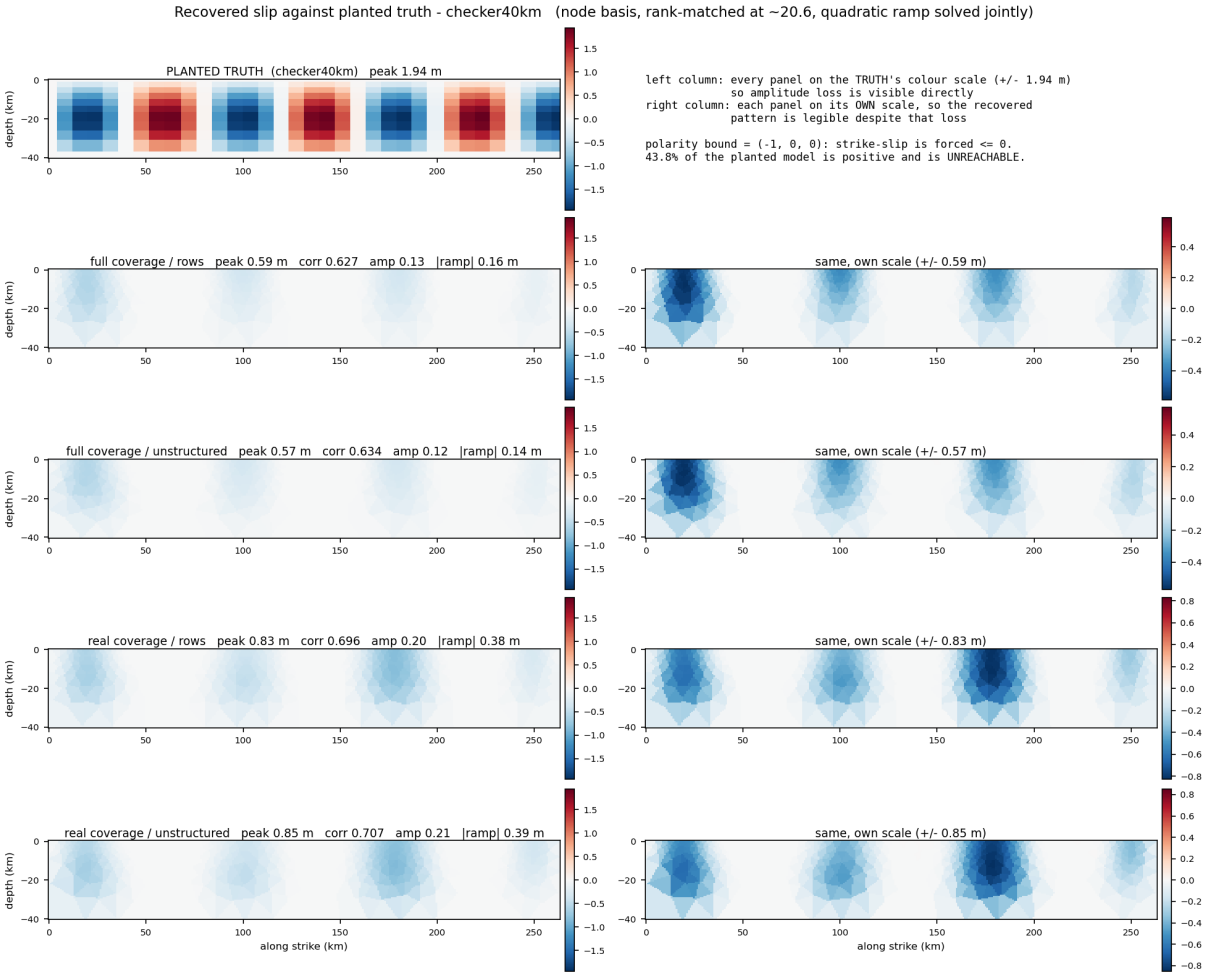


Figure 2: Planted sign-alternating checkerboard (top) against four recovered models. Left column shares the truth's colour scale; right column is autoscaled per panel. Every recovered model is single-signed — the polarity constraint forbids the positive lobes, so the positive cells are returned as exactly zero. This is the figure that identified the test as invalid.

rank	λ	correlation	rel. error	amplitude	max |ramp| (m)
10	381.0	0.275 ± 0.014	0.618 ± 0.002	0.904	0.667
21	142.5	0.438 ± 0.024	0.494 ± 0.005	0.950	0.411
39	51.4	0.619 ± 0.019	0.382 ± 0.007	1.082	0.290
79	**12.8**	$\mathbf{0.715 \pm 0.022}$	$\mathbf{0.353 \pm 0.015}$	1.287	0.263
129	3.8	0.692 ± 0.016	0.392 ± 0.025	1.416	0.269
198	1.1	0.519 ± 0.047	0.609 ± 0.043	1.878	0.271

Table 4: Five noise realisations of the 40 km magnitude checkerboard, real coverage, unstructured layout. Uncertainties are the seed spread. Every solve converged.

The error curve turns over at effective rank 79 — 3.8 times the L-curve corner — and the corner is 40% worse. The seed spread is 0.015 at the optimum against a 0.14 gap to the corner, so this is a property of the problem and not of one realisation.

The three panels of Figure 3 separate the two error terms cleanly. Pattern correlation peaks at the same rank the total error minimises. Amplitude ratio, by contrast, climbs *monotonically* through 1.0 and reaches 1.88 at rank 198 — that is the variance term, and it is why amplitude ratio must never be read as a recovery statistic on its own.

This does not make the L-curve wrong; it makes it a different criterion. The L-curve reads the bend in misfit against roughness and needs no truth, which is exactly why it is used. What this measures is that on this problem the bend sits at about a quarter of the rank the data can actually carry. Two caveats bound the claim: it is measured on a 40 km checkerboard, and a target with different spectral content may optimise elsewhere; and the ramp is solved jointly, which absorbs much of the correlated noise (max |ramp| is roughly flat at 0.26–0.29 m from rank 39 upward, against 0.07 m with no noise) and so flattens the variance penalty that would otherwise punish low damping sooner.

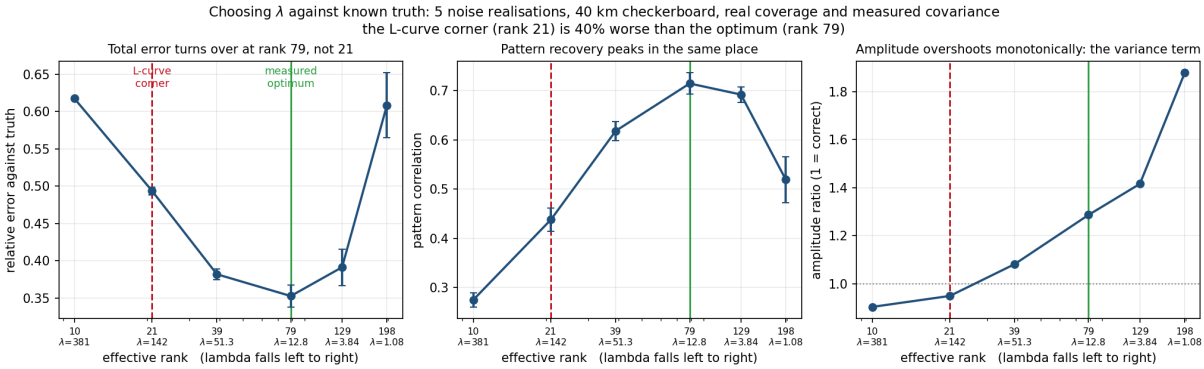


Figure 3: Choosing λ against known truth. Left: total error against truth, minimised at rank 79. Centre: pattern correlation, peaking in the same place. Right: amplitude ratio, climbing monotonically past 1.0 — the variance term made visible. Error bars are the spread over five noise realisations.

7.10 Two things that do not help: more elements, and more data

The 20 km target fails at every damping. Two candidate explanations were tested directly and both are refuted, which leaves an information limit.

Doubling the element budget changes nothing. At 660 elements the correlations across the ladder are 0.018 / 0.034 / 0.070 / 0.169; at 1320 they are 0.018 / 0.035 / 0.064 / 0.156 — indistinguishable, and marginally worse where it matters. Consistent with the discretisation ceiling, which already put 99.3% of that target's amplitude inside the coarser mesh: the extra parameters land in the null space, because effective rank is set by the data.

Full coverage does not help either, and at 40 km it hurts. Removing every coverage mask raises the observation count from about 7 200 to about 17 200 and moves the 20 km correlations by less than 0.002 at the first three rungs. At 40 km the effect is *negative* and reproduces on both layouts:

full minus real coverage	rank 10	rank 21	rank 40	rank 79
40 km, unstructured	+0.003	−0.043	−0.053	−0.019
40 km, rows	−0.009	−0.040	−0.046	−0.015
20 km, unstructured	+0.000	+0.002	+0.001	−0.021

The mechanism is the comparison itself: at *matched effective rank* the same rank budget is spread over 2.4× as many observations, so each is damped harder. The operational consequence is worth stating plainly — **more data does not buy resolution unless λ is relaxed to spend it**, which is the same lesson as Section 7.9 arriving from the other direction.

7.11 Summary: λ dominates

Four independent lines of evidence say the regularisation weight determines the answer and the other design choices are second-order:

1. Four mesh resolutions at matched effective rank agree on rms misfit to 0.3% and on M_w to 0.005.
2. Four mesh designs on the real six-scene data gave variance reduction 99.65–99.67% and M_w 7.594–7.607.
3. Two layouts, scored against known truth, agree to 1%.
4. Recovery of a fixed target varies from correlation 0.28 to 0.73 across the rank ladder alone, while swapping the mesh layout at fixed rank moves it by 0.015 and changing the target wavelength from 40 to 60 km moves it by 0.02 (Table 3).

The fourth is the sharpest: on the same data, λ moves the answer roughly **thirty times** as far as the mesh layout does. And Section 7.9 turns that from a ranking into a number — the operating point everything else was measured at was itself **40% worse than the best available**, and no mesh, layout, budget or coverage change recovers anything like that much.

Ortega-Culaciati et al. (2021) reach the same conclusion from theory and respond by fixing a fine mesh and varying the prior instead. Our data supports that choice. The honest headline is that the joint loop is an *efficiency* method, that the adaptive quadrature is a genuine and substantial computational result, and that the model itself is set by the prior.

8 Bugs worth recording

Each of these produced plausible output while being wrong, which is the only reason they are worth listing.

1. **LOS sign inverted** (Section 2.1). Every LOS product mirrored. Found by a dextral fault recovering sinistral slip, not by a test; the tests were self-consistent under a global flip.
2. **Smoothing operator units** (Section 3.2). Cotangent weights mixed with mean-value weights, a factor of the median edge length (2397) apart. Symptom: one patch of anomalous slip.
3. **Sandwich covariance** (Section 6.1). Weight matrix applied once instead of twice; 28.78× too large, correlation against Monte Carlo unchanged.
4. **Bias evaluated at a moving reference** (Section 6.2). Produced a bias curve with the wrong sign of slope.
5. **Checkerboard against a polarity bound** (Section 7.8). An entire recovery suite measuring its own constraint.
6. **Polarity diagnostic inverted.** The check written to catch (5) tested the reachable set instead of the forbidden one, reporting "100% forbidden" for a target lying entirely inside the bound. Diagnostic only; no score affected.

7. **Coverage mask reached the forward model through the look vectors,** not through the LOS field — so a "full coverage" synthetic world was byte-identical to the masked one, both at 24% coverage, until the look vectors were filled too.

9 Open items

- **Gaussian-fit kernel width.** The one variant of the published metric that has never been implemented, and the one the diagnosis in Section 5 points at.
- **The near-field quadrature cap** (Section 2.3) is a known tolerance violation inside $r \approx 2L$. Bounded and reported, not fixed.
- **Multi-fault geometry.** Blocked on the boundary-node mask, which tests global extrema, so a second fault's ends silently lose their zero-slip rows. The fix is a per-component label; everything downstream falls out unchanged.
- **The A/B loop comparison predates the smoothing-operator fix** and has not been re-run.
- **Effect of the mesh on where slip goes at depth.** The layouts agree on scalars and on truth-referenced error, but 64% of their mutual disagreement sits below 20 km depth, where $\mathrm{diag}(\mathbf{R}) \approx 0$. Neither can be called right there; the honest statement is that the data does not constrain it.

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